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Section	BSSE 6
Subject	Software Project Management

LAB 7

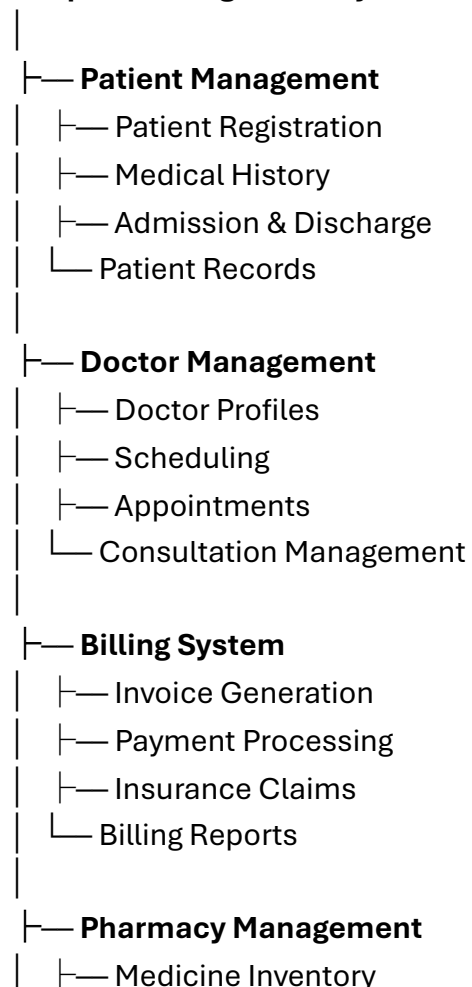
Scenario 1 — Hospital Management System

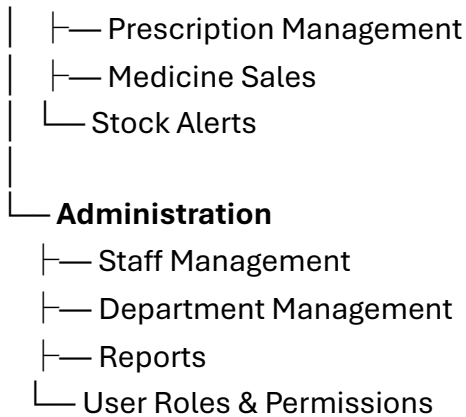
System Overview

A Hospital Management System is developed to digitally manage hospital operations including patients, doctors, billing, pharmacy, and administration. The system improves efficiency, record management, and communication between departments.

1. Tree Structure (Text-Based WBS)

Hospital Management System





2. Short Explanation

The Hospital Management System is divided into five major modules.

Patient Management handles registration and records.

Doctor Management handles appointments and schedules.

Billing System processes invoices and payments.

Pharmacy Management controls medicines and prescriptions.

Administration manages staff and departments.

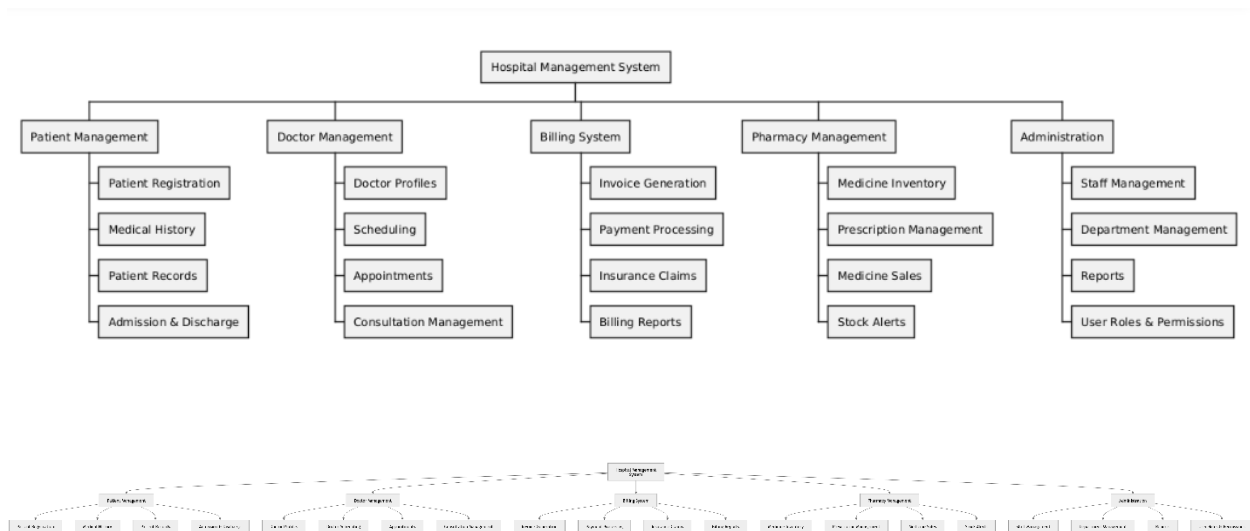
This WBS helps divide the project into manageable work packages for planning and development.

3. Deliverables

Module	Deliverable
Patient Management	Patient Records System
Doctor Management	Appointment Scheduling
Billing System	Invoice and Payment Module
Pharmacy Management	Medicine Inventory System
Administration	Staff & Department Control

4. Work Packages

Work Packages	Assigned To	Estimated Time
Patient Registration	Backend Developer	3 Days
Appointment Scheduling	Frontend Developer	4 Days
Invoice Generation	Finance Module Team	3 Days
Medicine Inventory	Database Team	5 Days
Staff Reports	Admin Team	2 Days



Scenario 2 — University Management System

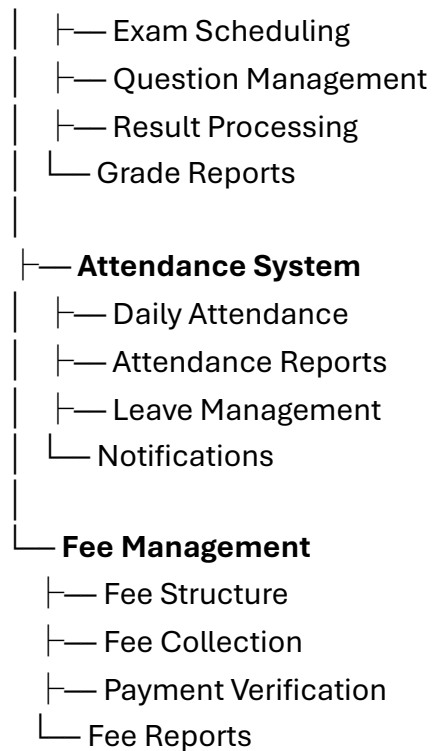
System Overview

A University Management System is developed to manage students, courses, examinations, attendance, and fee processing digitally.

1. Tree Structure (Text-Based WBS)

University Management System

- |
- |— Student Management
 - |— Student Registration
 - |— Student Profiles
 - |— Enrollment Management
 - |— Student Records
- |— **Course Management**
 - |— Course Creation
 - |— Teacher Assignment
 - |— Course Allocation
 - |— Course Material
- |— **Examination System**



2. Short Explanation

The University Management System is divided into modules that handle academic and administrative tasks.

Student Management controls student records and enrollment.

Course Management handles subjects and teachers.

Examination System manages exams and results.

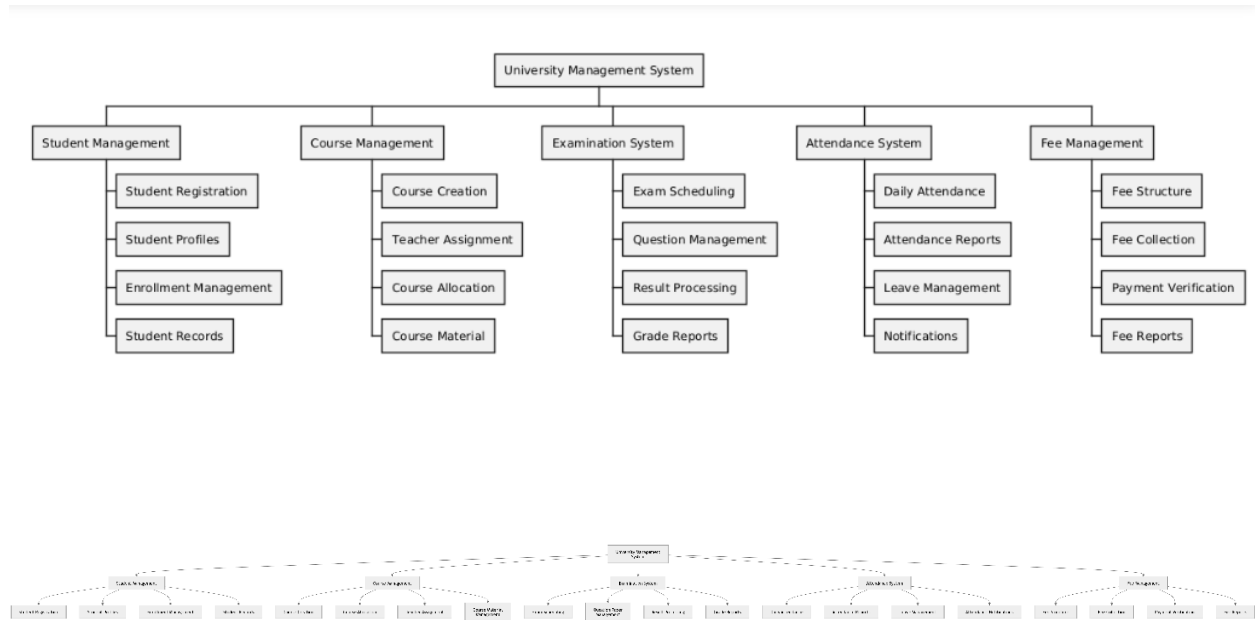
Attendance System tracks student attendance.

Fee Management handles student payments and reports.

The WBS ensures clear project planning and easier task allocation.

3. Deliverables

Module	Deliverable
Student Management	Student Information System
Course Management	Course Allocation Module
Examination System	Online Result System
Attendance System	Attendance Tracking
Fee Management	Fee Collection System



Scenario 3 — Online Food Delivery System

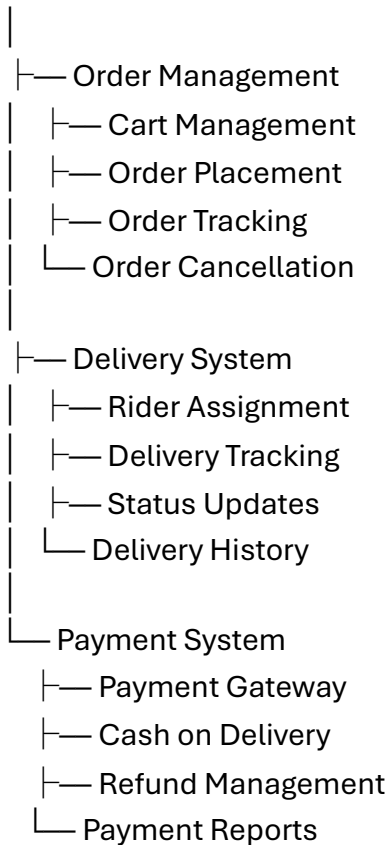
System Overview

An Online Food Delivery System is developed similar to FoodPanda or Uber Eats for managing customers, restaurants, orders, delivery, and payments.

1. Tree Structure (Text-Based WBS)

Online Food Delivery System

- |
- |— Customer Module
 - |— Customer Registration
 - |— Login & Profile
 - |— Browse Restaurants
 - |— Ratings & Reviews
- |
- |— Restaurant Module
 - |— Restaurant Registration
 - |— Menu Management
 - |— Order Acceptance
 - |— Restaurant Reports



2. Short Explanation

The Online Food Delivery System contains modules for customers, restaurants, orders, delivery, and payments.

Customers can browse restaurants and place orders.

Restaurants manage menus and orders.

The Delivery System tracks riders and deliveries.

The Payment System handles online transactions and refunds.

The WBS breaks the complete system into manageable tasks for development and monitoring.

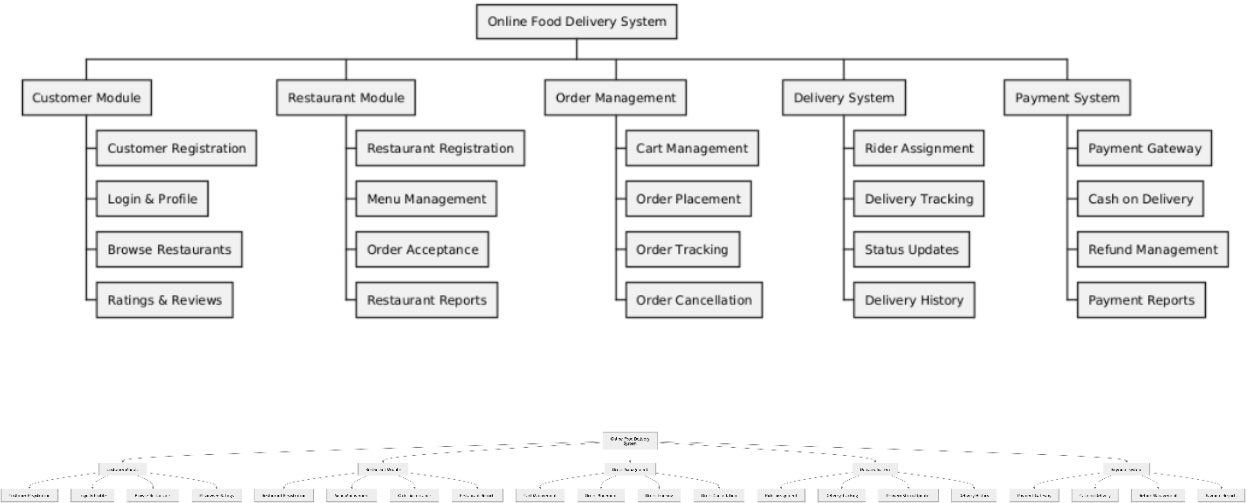
3. Deliverables

Module	Deliverable
Customer Module	Customer Application
Restaurant Module	Restaurant Dashboard
Order Management	Order Tracking System

Delivery System	Rider Tracking Module
Payment System	Secure Payment Gateway

4. Work Packages

Work Package	Assigned To	Estimated Time
Customer Registration	Frontend Team	2 Days
Menu Management	Backend Team	4 Days
Order Tracking	API Team	5 Days
Delivery Tracking	Mobile Team	4 Days
Payment Integration	Payment Team	6 Days



Smart Co-Working Space Booking & Management System

Complete Lab Data for WBS Submission

Based on your uploaded SPM project document.

1. Project Title

Smart Co-Working Space Booking & Management System

2. Problem Statement

Many co-working spaces still use manual booking methods such as spreadsheets, phone calls, and paper reservations. This creates:

- Double bookings
- Scheduling conflicts
- Poor workspace utilization
- Payment handling issues
- Time-consuming administrative work

An automated system is required to provide:

- Real-time workspace booking
 - Membership management
 - Secure online payments
 - Admin dashboard
 - Digital record management
-

3. Project Objectives

1. Automate booking and cancellation processes
 2. Provide real-time workspace availability
 3. Manage desks, cabins, and meeting rooms
 4. Support flexible membership plans
 5. Reduce booking conflicts
 6. Enable secure online payments
 7. Provide admin dashboard
 8. Generate reports and analytics
-

4. System Overview

The Smart Co-Working Space Booking & Management System is a web-based platform that allows users to:

- Check workspace availability
- Reserve workspaces online
- Manage memberships
- Make secure payments
- Monitor bookings and reports

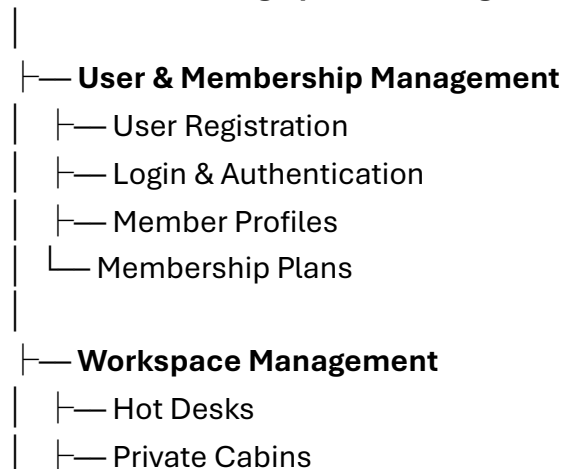
The system improves efficiency and reduces manual work.

5. Stakeholders

Stakeholder	Role
Co-Working Space Owner	Project Sponsor
Members / Clients	Book workspaces
Development Team	Develop system
Reception Staff	Handle walk-in bookings
System Administrator	Manage system
Payment Gateway Provider	Handle online payments

6. WBS Tree Structure (Text-Based)

Smart Co-Working Space Booking & Management System



└─ Meeting Rooms
└─ Availability Status
└─ Booking Management
└─ Check Availability
└─ Reserve Workspace
└─ Cancel Booking
└─ Prevent Double Booking
└─ Payment Management
└─ Online Payments
└─ Stripe / PayPal Integration
└─ Payment Records
└─ Refund Management
└─ Admin Dashboard
└─ Manage Spaces
└─ Manage Bookings
└─ Manage Members
└─ System Settings
└─ Reports & Monitoring
└─ Booking Reports
└─ Payment Reports
└─ Workspace Utilization
└─ Performance Monitoring

7. Deliverables

Module	Deliverable
User Management	Registration & Login System
Workspace Management	Workspace Availability System
Booking Management	Reservation Module
Payment Management	Secure Payment Gateway
Admin Dashboard	Management Panel
Reports & Monitoring	Analytics & Reports

8. Work Packages

Work Package	Assigned To	Estimated Time
User Registration	Frontend Developer	3 Days
Authentication System	Backend Developer	4 Days
Workspace Availability	Database Team	5 Days
Booking Module	Full-Stack Developer	6 Days
Payment Integration	Payment Team	5 Days
Admin Dashboard	Frontend Team	4 Days
Reporting Module	Analyst Team	3 Days

9. Project Phases

Phase Activities

Planning	Requirements gathering
Design	UI/UX and database design
Development	Frontend + Backend coding
Testing	Unit & integration testing
Deployment	Cloud hosting setup
Maintenance	Monitoring and updates

10. Functional Requirements

1. User registration and login
 2. Workspace booking system
 3. Real-time availability checking
 4. Membership management
 5. Online payment processing
 6. Booking cancellation
 7. Admin dashboard
 8. Report generation
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11. Non-Functional Requirements

Requirement	Details
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Security	JWT authentication & HTTPS
Performance	Response under 3 seconds
Availability	99% uptime
Scalability	Support multiple locations
Usability	User-friendly interface

12. Technologies Used

Technology	Purpose
React.js / Next.js	Frontend
Node.js / Express.js	Backend
MySQL / PostgreSQL	Database
Stripe / PayPal	Payment Gateway
AWS	Cloud Hosting
GitHub	Version Control

13. Short Explanation for Report

The Smart Co-Working Space Booking & Management System is divided into major modules such as User Management, Workspace Management, Booking Management, Payment Management, Admin Dashboard, and Reporting System.

Each module is further decomposed into smaller tasks and work packages. This Work Breakdown Structure helps in project planning, scheduling, resource allocation, and progress tracking. The hierarchical decomposition ensures better project management and reduces complexity during development.

